CDS6314 DATA MINING

GROUP 7

Title: Predict Future Trends In Mortality Rates Based On Historical Data And Current Risk Factors

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Abstract

Predicting trends in mortality rates is crucial for all the public health planning and resources allocation. The purpose of this project aims to predict these kinds of trends using historical data and the current risk. The dataset is sourced from Kaggle, which includes 34 attributes and 6121 instances. This dataset includes from multiple years and almost all countries in the world. The dataset will be used to apply the data mining techniques. The project encompasses data preprocessing, exploration, and mining, with tasks distributed among group members to ensure thorough analysis.

The methodology involves cleaning the dataset to look for any inconsistencies. The methodology also involves an in depth exploration to understand the distribution of death rates over time across the countries. The data mining methodology can be used to predict mortality trends. This prediction can help identify emerging patterns, potential areas of concern, and opportunities for new invitations, providing a better solution for all people.

The main objective will enhance the precision of predictions and provide valuable insights into the critical risk factors influencing mortality trends. The final deliverables will include a comprehensive report detailing the findings, Python code, and a presentation. By providing a reliable forecast of mortality trends, this project aims to support public health decision-making and policy formulation, ultimately contributing to better health outcomes on a global scale.

Introduction

Understanding and predicting mortality rates is a crucial component of public health planning and policy making. Assessing health outcomes by both mortality and morbidity provides a more encompassing view on health outcomes[1]. By analysing historical data and current risk factors, researchers and policymakers can identify patterns and anticipate future trends, enabling proactive measures to mitigate health risks and allocate resources more effectively. This research aims to build a robust predictive model using data mining techniques to uncover significant patterns and highlight areas of concern within mortality rates worldwide.

The motivation behind this study stems from the need to address the global burden of diseases and health conditions that contribute to mortality. With a comprehensive dataset encompassing a wide range of causes of death, such as meningitis, Alzheimer's disease, malaria, and cardiovascular diseases, this research seeks to leverage historical and current data to forecast future mortality trends. This predictive capability is particularly vital in a rapidly changing world where factors like emerging diseases, environmental changes, and shifting demographic patterns continuously alter health landscapes.

The primary objective of this research is to develop a predictive model that accurately forecasts mortality rates based on historical data and present-day risk factors. By employing advanced data mining techniques, we aim to identify underlying patterns and relationships within the data, which can provide valuable insights into the dynamics of mortality. This model will not only predict future trends but also help in pinpointing specific causes of concern, thereby guiding targeted interventions and policy decisions.

Through this comprehensive analysis, we aim to contribute to the global understanding of mortality trends, providing a foundation for improved public health strategies and resource allocation. The predictive model developed in this research has the potential to serve as a valuable tool for health professionals, policymakers, and researchers in their efforts to combat mortality and improve population health outcomes.

Literature Review

"Machine learning-based mortality rate prediction using optimised hyper-parameter" is written by Khan, Abbas and Truong (2020) [2]. The article discusses Gaussian Process Regression (GPR) for predicting mortality rates. GPR is a machine learning technique that can be used for regression and classification tasks. It is a non-parametric method that assumes the data follows a Gaussian distribution. In the context of mortality rate prediction, GPR can be used to model the relationship between risk factors and mortality rates. The model can then be used to predict future mortality rates based on new data.

The authors compare GPR to Polynomial Regression for mortality rate prediction. They find that GPR is more accurate than Polynomial Regression. This is likely because GPR is a more flexible method that can capture nonlinear relationships between variables. Polynomial Regression is limited to modelling linear relationships.
This paper employs classification techniques using ensemble methods, specifically Random Forest and Gradient Boosting, to predict mortality rates based on historical data and various risk factors. Hence, by taking these into consideration for this project, it could prove to be helpful in enhancing the precision of predictions and identifying critical risk factors influencing mortality trends.

Pourhomayoun and Shakibi (2021) made a research paper with the title "Predicting mortality risk in patients with COVID-19 using machine learning to help medical decision making" [3]. The research utilises neural networks, specifically Long Short-Term Memory (LSTM) networks, which are a type of recurrent neural network (RNN), for forecasting mortality rates. The use of LSTM networks significantly improved the accuracy of mortality predictions in COVID-19 patients. This can be critical for early intervention and resource allocation in healthcare settings. These techniques can help demonstrate how to integrate time-dependent patterns and leverage advanced neural network architectures to enhance the prediction accuracy of future mortality trends, which may be used in this project.

Furthermore, Moulaei, Shanbehzadeh, and Mohammadi-Taghiabad (2022) have also written a paper titled "Comparing machine learning algorithms for predicting COVID-19 mortality". In this paper, the researchers have applied deep learning models, particularly convolutional neural networks (CNNs) and recurrent neural networks (RNNs), for mortality prediction using electronic health records (EHRs). This research provides insights into advanced deep learning methodologies that can handle complex data structures, potentially leading to more nuanced and accurate mortality rate predictions. Therefore, deep learning can also be considered to detect more complex things which can help in predicting future mortality rates.

Lastly, the final related research paper is written by Chicco and Jurman (2020) and is titled "Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone". The study uses a variety of machine learning techniques, including ensemble methods, neural networks, and traditional regression models, to predict

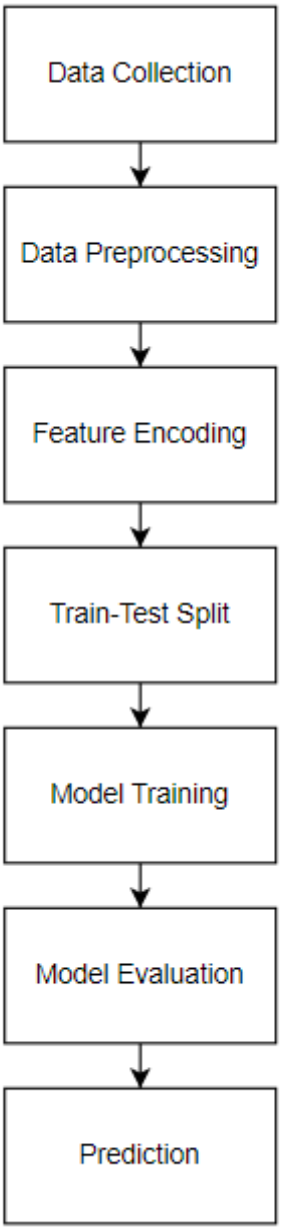
mortality rates from large-scale datasets. The research also highlights the significance of model validation and the use of cross-validation techniques to ensure the generalizability of the predictions. Hence, incorporating these practices into this project can help ensure that the models used are not only accurate but also reliable and generalizable to different populations.

Methodology

Overall Framework

The data mining pipeline includes the following steps: data preprocessing, exploratory data analysis (EDA), feature selection, data transformation, model training, and evaluation.

Project Framework



The data mining pipeline implemented for this project is designed to comprehensively analyse and predict mortality trends using historical data. The framework above is our overall

project. The dataset undergoes preprocessing to ensure the quality and consistency of the project. This will help by creating a clean environment to run all the necessary analysis.

Next, categorical features are encoded into numerical representations. This is really important for machine learning algorithms to effectively process the data. This allows valuable information to be contained within the categorical variables. After the dataset is prepared it will be splitted into two subsets, training and test set. The difference is that the training set is used to train a predictive model, allowing the model to learn the patterns in the data while the test set serves as an independent dataset for evaluating the model performance.

The model must then be trained after the data has been divided. Three types of models are used in this framework: random forest, XGBoost, and linear regression. The models learn the associations between the independent variables and the dependent variable by fitting them to the training data.

Performance measures like the average squared error and the R-squared score are used to assess each training model. These metrics give a sense of the model's prediction accuracy and quantify how well it matches the training set. It is essential to assess each model's performance in order to determine its efficacy and pinpoint any areas that require improvement.

Dataset

The dataset is from Kaggle (Cause of Deaths around the World). There are 34 attributes with 6121 instances in the dataset. The data is collected from global health organisations and databases for ensuring reliability and completeness.

The attributes with description are stated below:

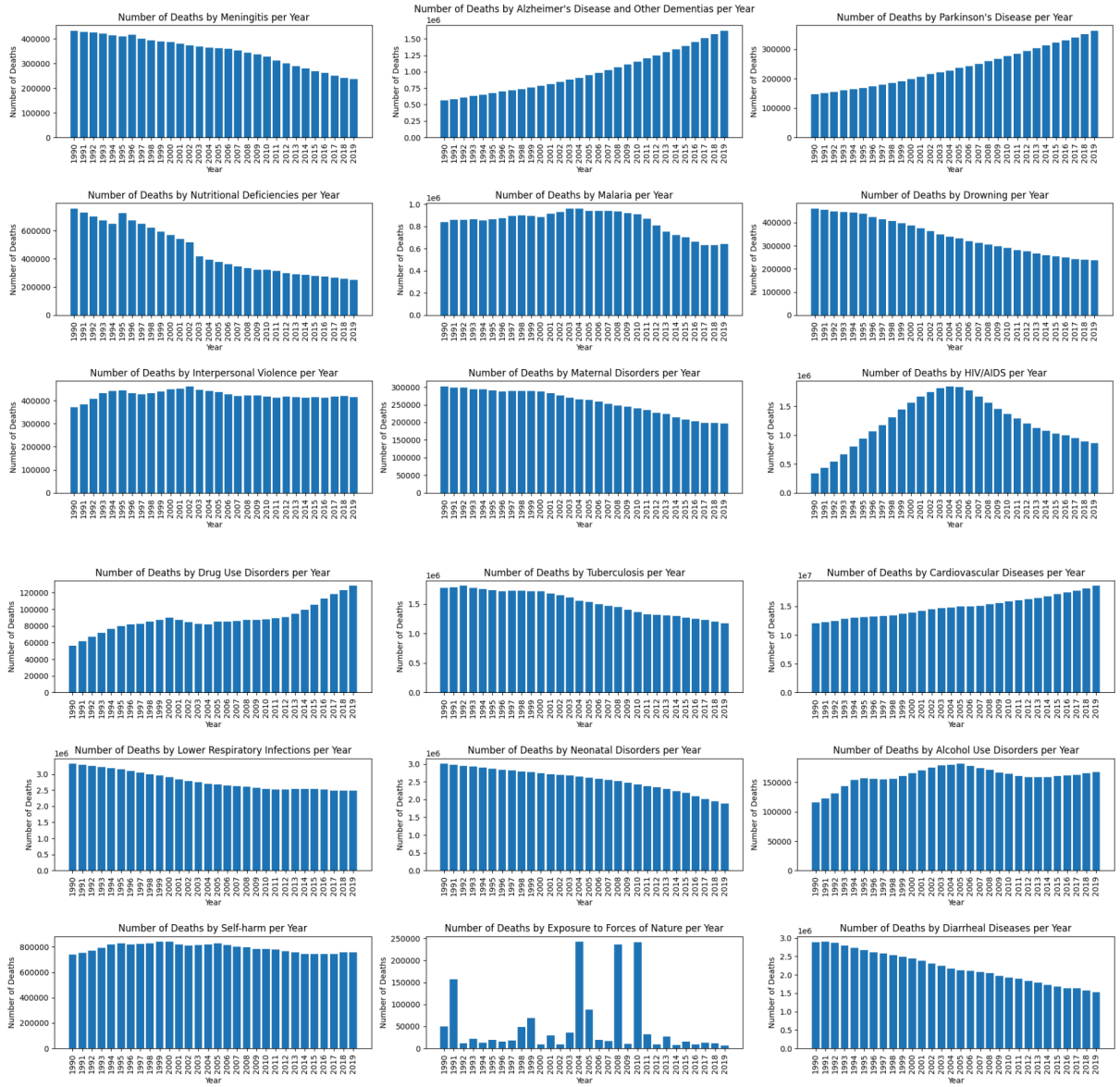
Attributes	Descriptions
Country/Territory	The country or territory where the data was recorded.
Code	The country or territory code.
Year	The year the data was recorded.
Meningitis	Number of deaths caused by meningitis.
Alzheimer's Disease and Other Dementias	Number of deaths caused by Alzheimer's disease and other dementias.
Parkinson's Disease	Number of deaths caused by Parkinson's disease.
Nutritional Deficiencies	Number of deaths caused by nutritional deficiencies.
Malaria	Number of deaths caused by malaria.
Drowning	Number of deaths caused by drowning.
Interpersonal Violence	Number of deaths caused by interpersonal violence.
Maternal Disorders	Number of deaths caused by maternal disorders.
HIV/AIDS	Number of deaths caused by HIV/AIDS.
Drug Use Disorders	Number of deaths caused by drug use disorders.
Tuberculosis	Number of deaths caused by tuberculosis.
Cardiovascular Diseases	Number of deaths caused by cardiovascular diseases.
Lower Respiratory Infections	Number of deaths caused by lower respiratory infections.

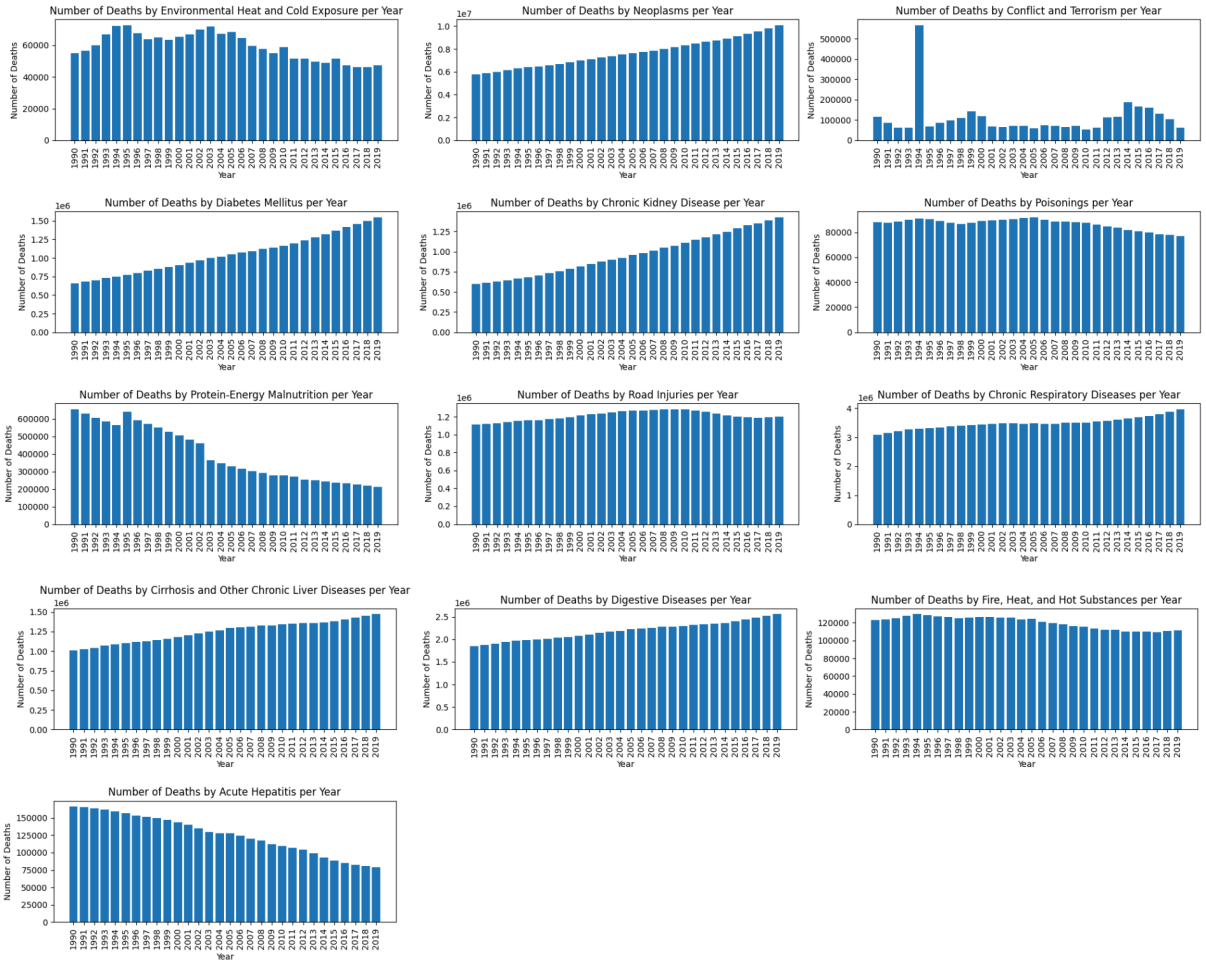
Neonatal Disorders	Number of deaths caused by neonatal disorders.
Alcohol Use Disorders	Number of deaths caused by alcohol use disorders.
Self-harm	Number of deaths caused by self-harm.
Exposure to Forces of Nature	Number of deaths caused by exposure to forces of nature.
Diarrheal Diseases	Number of deaths caused by diarrheal diseases.
Environmental Heat and Cold Exposure	Number of deaths caused by environmental heat and cold exposure.
Neoplasms	Number of deaths caused by neoplasms (cancers)
Conflict and Terrorism	Number of deaths caused by conflict and terrorism.
Diabetes Mellitus	Number of deaths caused by diabetes mellitus.
Chronic Kidney Disease	Number of deaths caused by chronic kidney disease.
Poisonings	Number of deaths caused by poisonings.
Protein-Energy Malnutrition	Number of deaths caused by protein-energy malnutrition.
Road Injuries	Number of deaths caused by road injuries.
Chronic Respiratory Diseases	Number of deaths caused by chronic respiratory diseases.
Cirrhosis and Other Chronic Liver Diseases	Number of deaths caused by cirrhosis and other chronic liver diseases.
Digestive Diseases	Number of deaths caused by digestive diseases.
Fire, Heat, and Hot Substances	Number of deaths caused by fire, heat, and hot substances.
Acute Hepatitis	Number of deaths caused by acute hepatitis.

Data Preprocessing

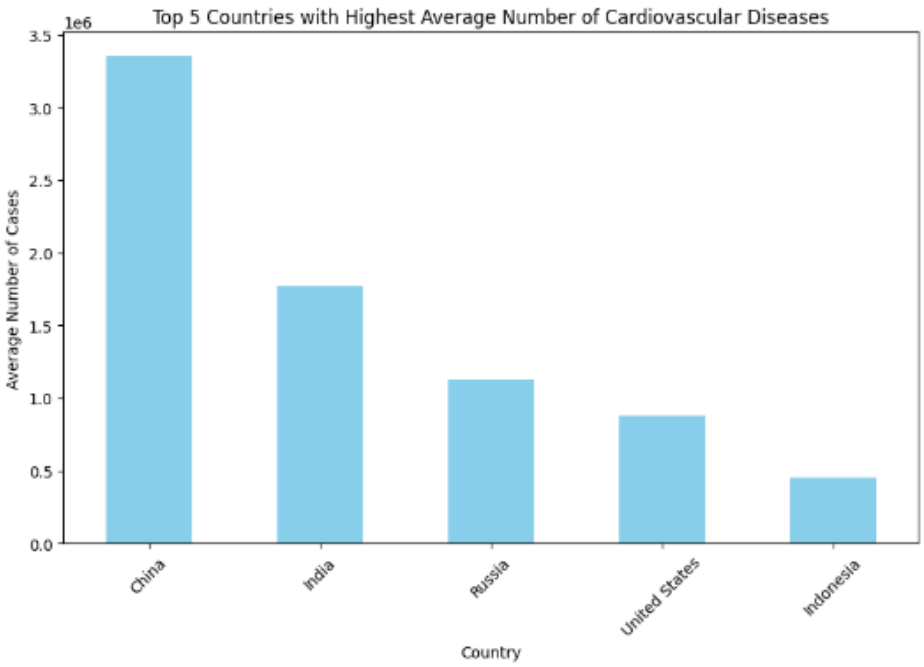
DataFrame is taken to create a new table summarising information about its columns. It shows the column names, the number of non-null values in each column for indicating missing data, and the data types of the columns. Then, the missing value is checked in the dataset and there is no missing value in the dataset.

Separate bar charts for each cause of death are used to visualise the number of deaths for each cause across the years included in the DataFrame. Besides, We calculate the correlations between numerical columns in the DataFrame and then visualise those correlations using a colour-coded heatmap. This helps identify potential relationships between different causes of death.

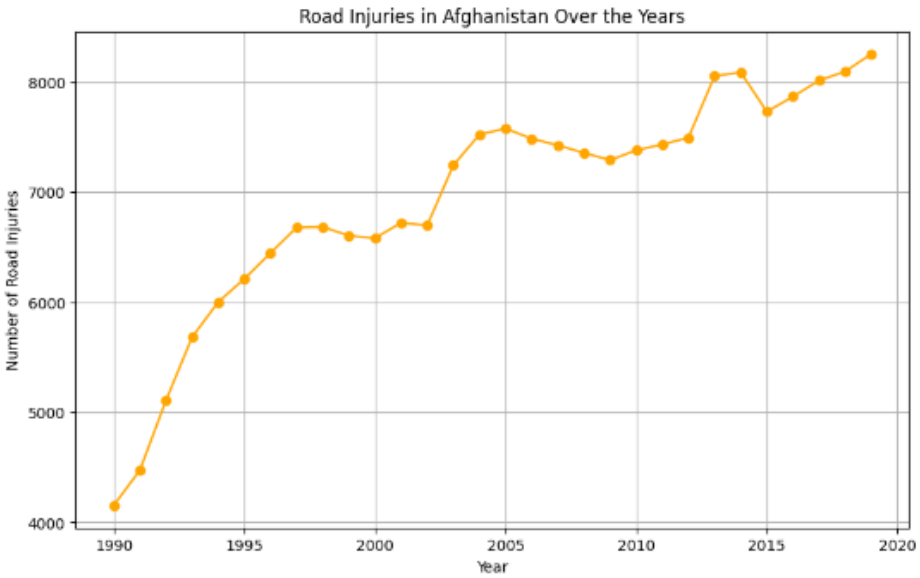



Number of Deaths by Environmental Heat and Cold Exposure per Year
Number of Deaths by Neoplasms per Year
Number of Deaths by Conflict and Terrorism per Year
Number of Deaths by Diabetes Mellitus per Year
Number of Deaths by Chronic Kidney Disease per Year
Number of Deaths by Poisonings per Year
Number of Deaths by Protein-Energy Malnutrition per Year
Number of Deaths by Road Injuries per Year
Number of Deaths by Chronic Respiratory Diseases per Year
Number of Deaths by Cirrhosis and Other Chronic Liver Diseases per Year
Number of Deaths by Digestive Diseases per Year
Number of Deaths by Fire, Heat, and Hot Substances per Year
Number of Deaths by Acute Hepatitis per Year
Number of Deaths
Year

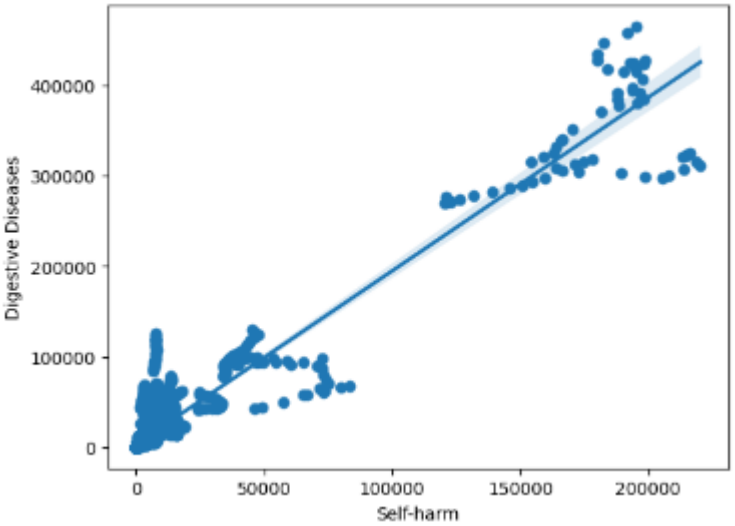
The distribution of various causes of death over time and across countries is aligned in the analysis. One of the questions is 'Top 5 Countries with the Highest Average Number of Cardiovascular Diseases'. Based on this question we know that China has the highest number of cardiovascular diseases while Indonesia has the lowest number of people with cardiovascular diseases.



Next question is 'Number of road injuries over the years in Afghanistan' as the year goes on the number of deaths increases. This might be caused due to the number of vehicles on the road increasing. The number of road injuries in the year 1990 was just above 4000 while the number of road injuries in 2015 was more than 7500.



‘Cause of death with the highest total occurrences across all countries and years’ based on this question, the conclusion is that the number of deaths by cardiovascular diseases is the highest across all the countries. Following this is neoplasm which is the second highest cause of death across all the countries. Finally, ‘Is there a correlation between self harm and the prevalence of digestive disease?’ The correlation between self harm and prevalence of digestive disease is that people with prevalence of digestive disease are more compared to people who do self harm.



Feature selection has been done by preparing the data for modelling by calculating the total deaths per year and separating the year and cause of death information (features) from the total deaths (target variable).

Data normalisation is performed to ensure features are on a similar scale. This prevents features with larger ranges from dominating the model and improves the performance and accuracy of the model in predicting future mortality rates.

Data Mining

In this project, we have decided to try 3 different models that can be used to make the predictions of future trends of death rates. The models used are Random Forest, Linear Regression and XGBoost. The main purpose of using these 3 different models is to find out which model is the most suitable to make predictions.

Firstly, the Random Forest model was chosen for its ability to handle non-linear relationships. We used 100 trees and evaluated its performance. The Random Forest model builds multiple decision trees using random subsets of the data and features. Each tree provides a prediction, and the final output is the average of these predictions. We have chosen this model because it can capture complex relationships in the data and provide more accurate predictions for trends in death rates. It is particularly useful when the relationship between features and the target variable is not strictly linear.

Next, Linear Regression was used as a baseline model to identify linear relationships in the data. We trained the model using the training data and evaluated its performance on the test data. We have chosen Linear Regression because it can provide a quick and straightforward baseline model for understanding the general trends in the data. It helps in identifying whether there is a linear relationship between time (years) and death rates.

Lastly, XGBoost was selected for its high accuracy and robustness. The model was trained with default parameters and evaluated using the test data. XGBoost builds decision trees sequentially, with each new tree attempting to correct the residual errors made by the previous trees. It uses a gradient descent algorithm to minimise the loss function, which measures the difference between the actual and predicted values. Hence, it is well-suited for handling complex datasets with intricate relationships between features. It can provide highly accurate predictions for death rates, capturing both linear and non-linear trends over time.

Parameters used for Random Forest:

```
# Training the Random Forest model
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
```

- **n_estimators=100**: The number of trees in the forest.
- **random_state=42**: Controls the randomness of the estimator, ensuring reproducibility.

Parameters used for Linear Regression:

```
# Training the linear regression model
model = LinearRegression()
model.fit(X_train, y_train)
```

In this implementation, we used the default parameters provided by the LinearRegression class from the scikit-learn library.

Parameters used for XGBoost:

```
# Training and fitting the XGBoost model
model = XGBRegressor(random_state=42)
model.fit(X_train, y_train)
```

- **random_state=42**: Ensures reproducibility of the results.

Evaluation

In this section, we evaluate the performance of three predictive models: Linear Regression, Random Forest, and XGBoost. The evaluation process aims to determine the effectiveness of each model in predicting future trends in death rates. By assessing these models, we can identify which model provides the most accurate and reliable predictions.

Evaluation Metrics

The models were evaluated using three metrics: **Mean Squared Error (MSE)**, **Root Mean Squared Error (RMSE)**, and **R^2 Score**.

Mean Squared Error (MSE) represents the average of the squared difference between the original and predicted values in the data set. It measures the variance of the residuals.

Root Mean Squared Error (RMSE) is the square root of Mean Squared error. It measures the standard deviation of residuals.

R-squared (R^2 Score) is a number that tells you how well the independent variable(s) in a statistical model explain the variation in the dependent variable. It ranges from 0 to 1, where 1 indicates a perfect fit of the model to the data.

Evaluation Metrics for Random Forest model:

```
Model Evaluation Results:
--------------------------
Mean Squared Error (MSE): 492393425.30526114
Root Mean Squared Error (RMSE): 22189.939731897903
R^2 Score: 0.9991605631879467
```

Evaluation Metrics for Linear Regression model:

```
Model Evaluation Results:
--------------------------
Mean Squared Error (MSE): 1.2218443343939465e-18
Root Mean Squared Error (RMSE): 1.105370677372051e-09
R^2 Score: 1.0
```

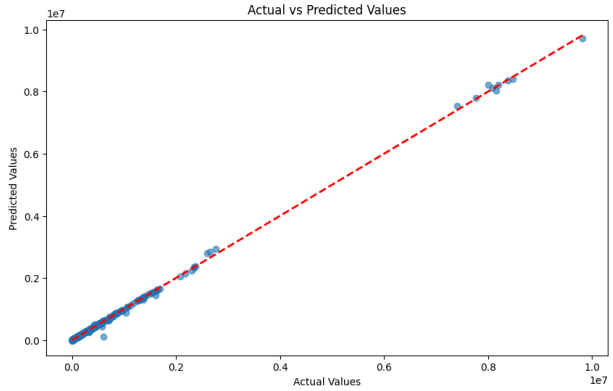
Evaluation Metrics for XG Boost model:

```
Model Evaluation Results:
--------------------------
Mean Squared Error (MSE): 518870761.63116354
Root Mean Squared Error (RMSE): 22778.73485580715
R^2 Score: 0.9991154243829691
```

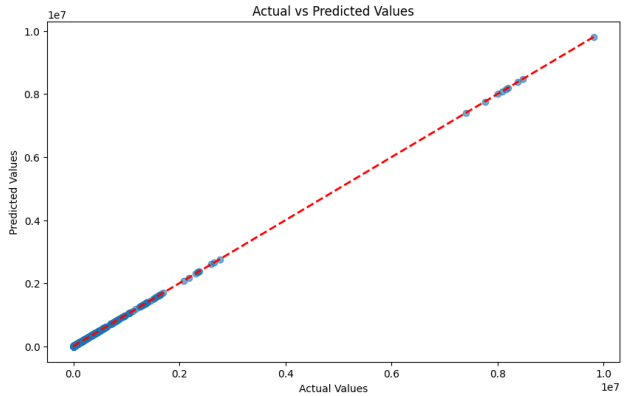
Results and Discussion

Actual vs Predicted

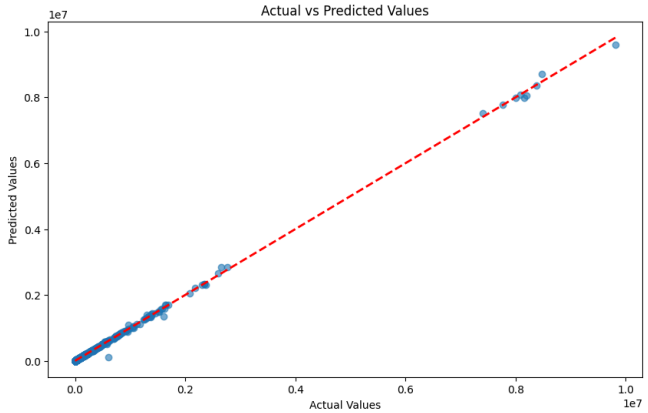
Random Forest



Linear Regression



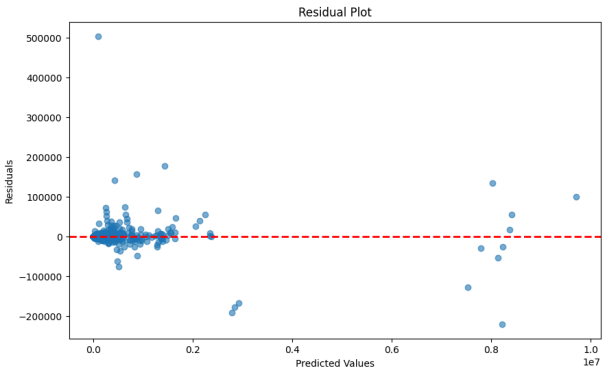
XG Boost



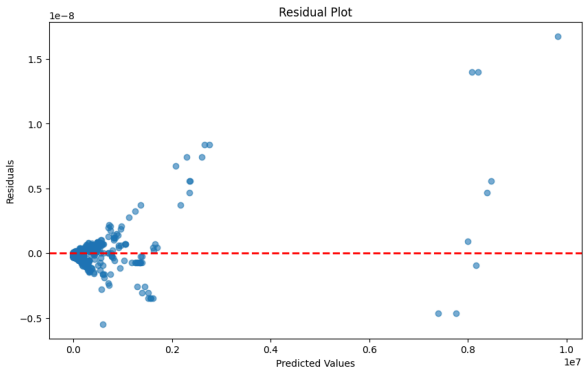
The graphs represent the actual vs predicted values in three models (Random Forest, Linear Regression, XGBoost). According to the graph most of the plots are close to the ideal line, meaning that the linear regression model's predictions are generally accurate and align well with the actual values without extreme outlier.

Residual plot

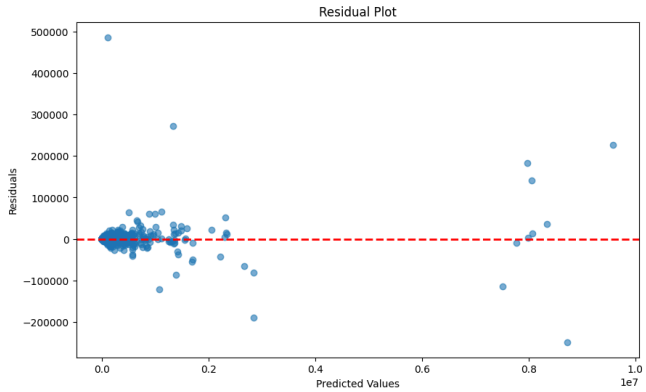
Random Forest



Linear Regression



XGBoost

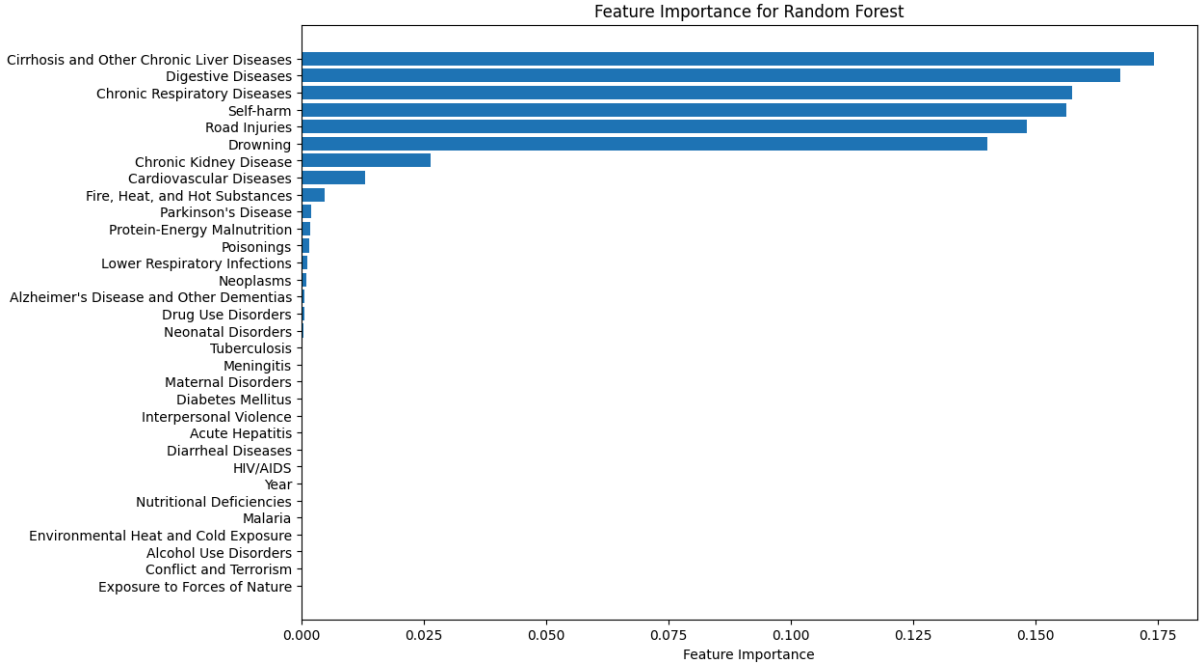


Most of the residuals are clustered near the zero line, indicating that the model's predictions are close to the actual values for a large portion of the data. There are some points, particularly on the right side of the plot, that have larger residuals, indicating that the model's predictions are less accurate for these data points. There are a few noticeable outliers with very high residuals, both positive and negative. These outliers could be points where the model significantly under or overestimated the actual values.

The clustering of points around the zero line suggests that the model performs reasonably well for a significant portion of the data. However, the spread and presence of outliers indicate areas where the model may not be performing optimally.

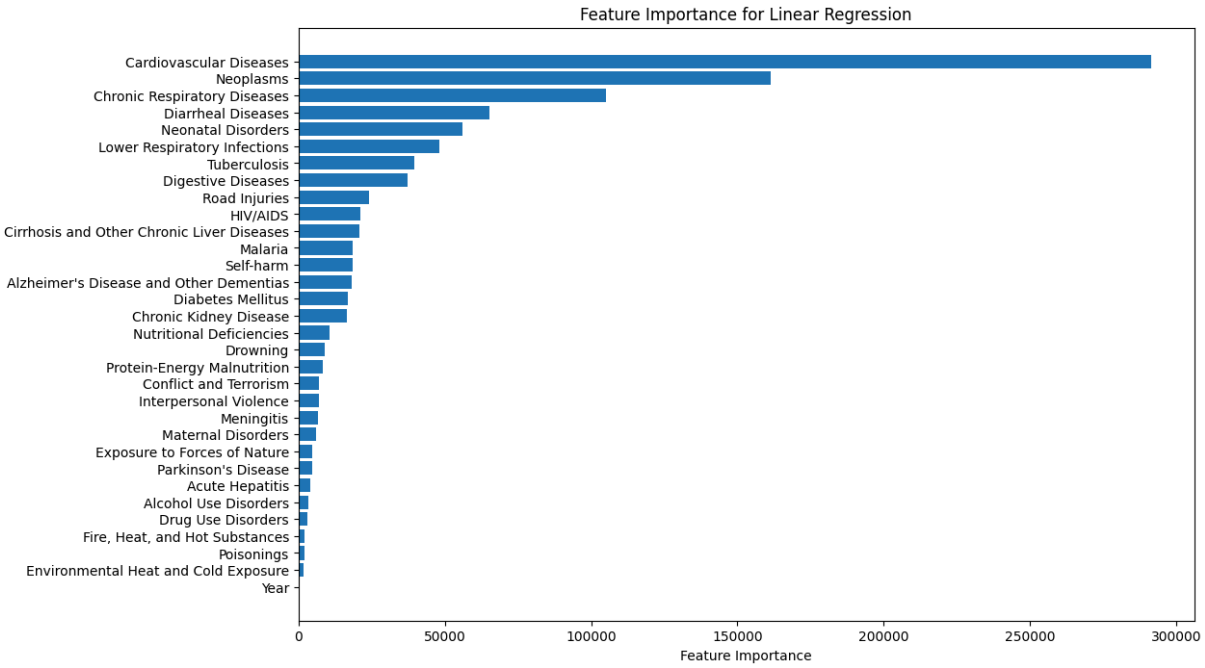
Feature Importance

Random Forest



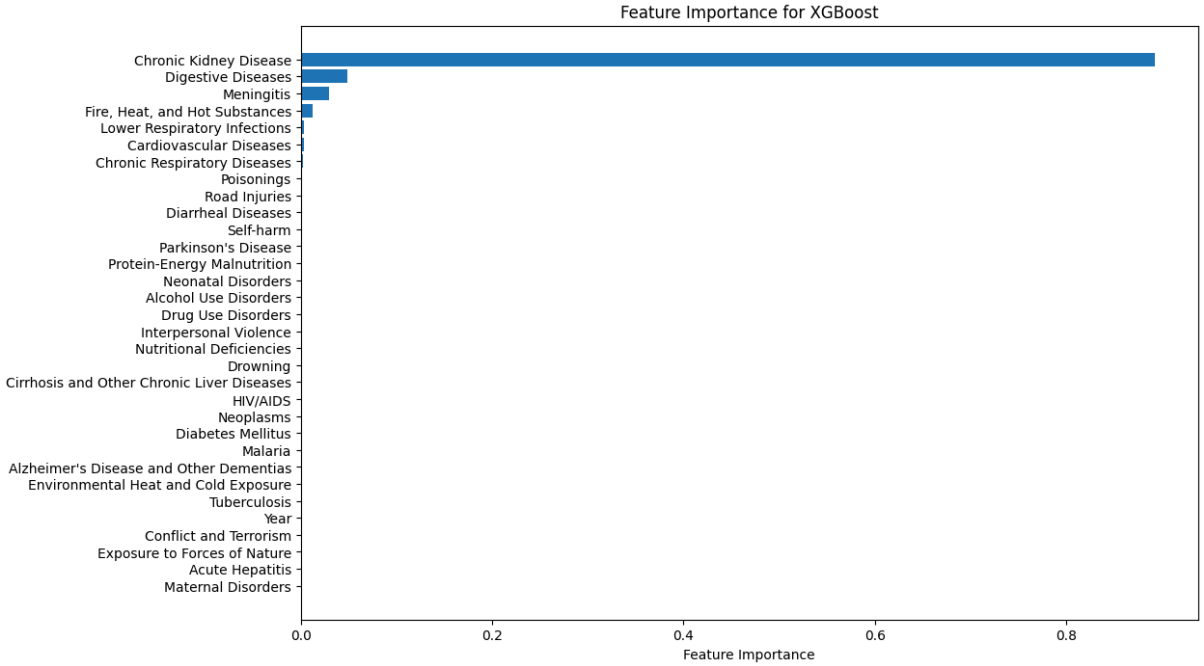
This is the feature importance graph for the Random Forest model. The ‘Cirrhosis and Other Chronic Liver Diseases’ feature stands out as the most influential, followed closely by ‘Digestive Diseases’, ‘Chronic Respiratory Diseases’, ‘Self-harm’. While, features such as 'Alzheimer's Disease and Other Dementias', 'Drug Use Disorders', and 'Neonatal Disorders' are the bottom 3 in influencing the prediction.

Linear Regression



This is the feature importance graph for the Linear Regression model. ‘Cardiovascular Diseases stand out the most when compared to others followed by ‘Chronic Respiratory Diseases’ and ‘Diarrheal Diseases'. 'Fire, Heat, and Hot Substances', 'Poisonings', and 'Environmental Heat and Cold Exposure' are the bottom 3 in influencing the prediction.

XGBoost



This is the feature importance graph for the XGBoost model. 'Chronic Kidney Disease' stands out as the most influential, followed by 'Digestive Diseases', and 'Meningitis'. These importance scores play a crucial role in guiding decision-making within the model, aiding in the prioritisation of features for accurate predictions. 'Fire, Heat, and Hot Substances', 'Lower Respiratory Infections', 'Cardiovascular Diseases', and ‘Chronic Respiratory Diseases’ are the bottom 4 in influencing the prediction.

Deployment

For this project we used Streamlit. Streamlit is an open-source app framework tailored for Machine Learning and Data Science teams, allowing them to create interactive web apps with ease. This particular application integrates Python's powerful data science libraries, making it an excellent choice for developing custom dashboards and interactive data visualisations.

The first thing the program does is load a dataset and show it to the user right away. Important data features are emphasised, offering a strong basis for additional analysis. Examples of these features are missing values, duplicate entries, and basic descriptive statistics. Determining any problems and comprehending the structure of the dataset depend on this preliminary data investigation.

The capability of this Streamlit program to carry out extensive exploratory data analysis (EDA) is one of its primary features. It creates visuals, such as bar charts that illustrate the yearly death toll from different causes, to demonstrate the distribution and patterns of data over time. The program also lists the top nations with the greatest average prevalence of particular illnesses. These statistical and visual summaries help users understand intricate data patterns by providing insightful information about the dataset.

Additionally, interactive charts are supported by the program, which improves user engagement and data understanding. It enables users to train models like XGBoost, Random Forest, and Linear Regression utilising the dataset by incorporating machine learning capabilities. Metrics like the R2 score and Root Mean Squared Error (RMSE) are used to assess the performance of the model when the data is divided into training and testing sets. With the help of this feature, users may create and evaluate prediction models right within the app.

This Streamlit application is a powerful tool for data analysis and machine learning, offering extensive features for data visualisation, model training, and evaluation. Its flexible deployment options make it suitable for a wide range of projects, from simple dashboards to complex, scalable applications.

Conclusion

Overall Findings

In this project, we explored the prediction of future trends in death rates using three different machine learning models: Random Forest, Linear Regression, and XGBoost. Each model was trained and evaluated on a dataset containing various causes of death, with the goal of identifying which model provides the most accurate predictions.

Firstly, the Random Forest model demonstrated a strong ability to capture non-linear relationships and interactions between features. It achieved a high R^2 Score, indicating excellent explanatory power. However, the high RMSE value suggested that individual predictions could still be significantly off, indicating room for improvement.

On the other hand, the Linear Regression model provided a perfect fit to the training data. However, such a perfect fit may also indicate overfitting, where the model performs extremely well on the training data but may not generalise well to unseen data.

In addition, the XGBoost model had results that outperformed both of the other models in terms of accuracy. Although the R^2 Score in this model is slightly lower compared to Random Forest, the RMSE value is also slightly higher. This indicates that while XGBoost captured the variance in the data well, it still had significant individual prediction errors.

Potential Use Cases or Importance

There are many insights that can be gained from this project, which can benefit public health planning and policy-making. There are also a few potential use cases that can be identified from the models that we have used. For example, forecasting future death rates and trends for specific causes of death. Additionally, identifying high-risk areas or populations that may require more intensive health interventions. Furthermore, supporting the development of health policies and programs aimed at reducing mortality.

Future Directions of the Work

While the models developed in this project provide valuable insights, there are several areas for future work to enhance their accuracy and applicability.

First of all, outside of these 3 models that were used, there are other advanced models which can be more suitable for predicting the future trends of death rates. For example, using neural network models such as Long Short-Term Memory (LSTM) and Autoencoders. These models might help in providing better predictive accuracy and applicability.

Aside from that, models such as the Random Forest and XGBoost can be further improved by further tuning the hyperparameters so that they could potentially reduce prediction errors such as the Root Mean Squared Error (RMSE) value for each model.

Lastly, to prevent overfitting in models such as the Linear Regression, implementing cross-validation techniques more extensively may also be of help. This can also help improve the models' robustness.

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